Bank ESG Performance and Lending Behavior in the Mortgage Market *

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Abstract

This paper investigates whether banks' environmental, social, and governance (ESG) ratings influence their mortgage lending decisions toward minority borrowers. Initial OLS estimates suggest that banks with higher ESG ratings issue fewer mortgage loans—in both volume and number—to minority applicants. The distribution of lending across racial groups is heterogeneous: high-ESG banks lend more to Asian applicants but significantly less to American Indian, Black, and Native Hawaiian borrowers. However, these patterns may be driven by omitted variables or reverse causality. To address endogeneity concerns, I employ an instrumental variable (IV) approach, using the political leanings of the counties in which bank branches are located as an instrument. The IV results indicate that banks' ESG practices have no causal impact on mortgage lending decisions. I propose an "irrelevance mechanism" to explain this finding, arguing that mortgage lending decisions are largely determined by the evaluation outputs of Automated Underwriting Systems (AUS) provided by Government-Sponsored Enterprises (GSEs), and that ESG metrics tend to prioritise internal workforce diversity over customer-facing lending practices.

Keywords: Algorithmic Approval;Banks; Corporate Finance; ESG; Mortgages

JEL classification: G21, G30, M14, R31

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1 Introduction

The rise of Environmental, Social, and Governance (ESG) criteria has fundamentally reshaped the corporate landscape. A central debate in modern finance is whether corporate investments in ESG reflect a genuine commitment to stakeholder value (Bénabou and Tirole (2010)) or represent an agency cost whereby managers engage in "greenwashing" for reputational benefits without altering core business practices. While a vast literature examines ESG in non-financial firms (e.g., Ferrell et al. (2016), Lins et al. (2017), Dyck et al. (2019)), its role in the banking sector is uniquely critical. Banks are the primary allocators of capital in the economy, and their lending decisions have profound distributive consequences. In recent years, US banks have made trillion-dollar public commitments to sustainable financing and social equity.[1] However, it remains an open question whether these high-level corporate ESG commitments actually translate into tangible changes in capital allocation at the micro level.

This paper investigates this question by examining the U.S. residential mortgage market—the largest consumer debt market in the world (Campbell (2006))—with a specific focus on credit provision to minority groups. Mortgage access is a primary driver of household wealth accumulation, yet there is a long-standing debate regarding racial disparities and discrimination in lending (Munnell et al. (1996); Boehm and Schlottmann (2007); Clarke and Rothenberg (2018); Ambrose et al. (2021); Bartlett et al. (2022)). Given the explicit focus of the "Social" pillar of ESG on diversity and inclusion, mortgage lending to minority groups provides a powerful setting in which to test whether banks' ESG ratings reflect substantive prosocial credit allocation or merely superficial corporate signaling.

Theoretically, the relationship between a bank's ESG rating and its lending to minority groups is highly ambiguous, presenting three competing hypotheses. Under the

[1]For example, in April 2021, JPMorgan Chase announced a commitment to finance and facilitate $2.5 trillion over 10 years to support climate action and sustainable development (see https://www.jpmorganchase.com/newsroom/press-releases/2021/jpmc-to-advance-climate-action-and-sustainable-dev-goals). In the same month, Bank of America announced a $1.5 trillion sustainable finance target by 2030, including a $1 trillion Environmental Business Initiative alongside commitments to socially inclusive development(see https://about.bankofamerica.com/en/making-an-impact/environmental-sustainability).

first hypothesis—the *prosocial signaling hypothesis*—banks respond to public enthusiasm for corporate responsibility by actively expanding credit access to minority groups. Because diversity and inclusion efforts are rewarded by ESG rating agencies, banks can use increased minority lending to signal their prosocial commitments and enhance their overall ESG ratings. However, this introduces a critical economic friction: if banks systematically relax their underwriting standards to maximise ESG ratings, irrespective of the borrowers' underlying credit conditions, they fundamentally increase the riskiness of their loan portfolios. While this behavior may benefit minority borrowers in the short term, the elevated default risk actively harms the interests of depositors, shareholders, and broader financial stability. In this scenario, the pursuit of ESG ratings explicitly distorts core business decisions.

Alternatively, under the *risk mitigation hypothesis*, high ESG ratings might perversely incentivize banks to lend less to minority groups. Due to historical and systemic economic disadvantages, minority applicants, on average, present lower credit scores and less accumulated wealth. If a bank strictly adheres to risk-based lending principles, it must restrict credit to riskier minority applicants. Furthermore, if a bank disregards these credit conditions to issue more loans, the inevitable increase in mortgage defaults will not only erode bank profitability but also invite negative regulatory scrutiny and public controversy. Because ESG rating agencies heavily penalize corporate controversies and governance failures, an increase in defaulted loans could severely damage a bank's ESG rating. Therefore, to protect both their profitability and their ESG ratings, highly rated banks have a strong incentive to "de-risk" their portfolios by actively reducing their credit exposure to disadvantaged groups.

A third possibility is the *irrelevance hypothesis*: a bank's ESG rating has no material relationship with its core mortgage lending decisions. In this scenario, high ESG ratings are achieved through highly visible, easily controllable corporate practices rather than fundamental changes in how the bank evaluates and approves routine mortgage applications. For example, rather than altering their aggregate risk appetite, banks may fulfill their social obligations by setting aside a fixed, predetermined budget for targeted philanthropic efforts or specialized, ring-fenced loan funds specifically dedicated to minority home buyers. Because these ESG initiatives are treated as distinct,

bounded corporate directives, the bank's ESG rating remains entirely decoupled from the vast majority of its day-to-day mortgage origination decisions.

To differentiate between these hypotheses, I conduct the empirical analysis by collecting ESG ratings for banks from Refinitiv, mortgage lending data released by the Federal Financial Institutions Examination Council (FFIEC) under the Home Mortgage Disclosure Act (HMDA) from 2009 to 2017, and financial information for banks from FR Y-9C. To merge these datasets, I use the CRSP-FRB link provided by the Federal Reserve Bank of New York, the CRSP/Compustat link provided by Wharton Research Data Services (WRDS), and HMDA lender file provided by Federal Reserve Bank of Philadelphia. I present two main findings. First, banks with high ESG ratings tend to issue fewer residential mortgage loans, both in terms of number and dollar amount. Second, the relationship differs across racial groups. Banks with high ESG ratings respond differently: they originate larger loan amounts to Asian applicants but less to American Indian, Black, and Native Hawaiian applicants.

Although the empirical evidence shows a negative relationship between banks' ESG ratings and their lending decision, it may be subject to a common identification challenge: low mortgage loan approval rates or reduced lending may be responding to the low mortgage loan demand from minority groups in addition to banks' ESG ratings. For instance, if demand for the mortgage loan decreases, banks with high ESG ratings also lend less to minority groups. To address endogeneity and assess causality, I implement an instrumental variable (IV) approach. I instrument bank's ESG ratings by using the average share of votes for the Democratic presidential candidate in the last presidential election across all counties where a bank's branches are located. This approach rests on evidence that branch staff in Democratic-leaning counties are more likely to support ESG-related activities (Di Giuli and Kostovetsky (2014)). Moreover, counties' election outcomes vary over time, offering plausibly exogenous time-series variation in banks' ESG performance. While the first-stage results confirm that this instrument strongly predicts bank's ESG performance, the second-stage two-stage least squares (2SLS) estimates show no causal relationship between a bank's ESG rating and its mortgage lending to minority groups.

Consistent with the 2SLS estimates, the empirical findings reject both the prosocial

signaling and risk mitigation hypotheses, providing strong empirical support for the irrelevance hypothesis. The negative correlation observed in the baseline OLS regressions is likely driven by unobservable local factors rather than deliberate bank policy. To explain exactly why a bank's ESG rating does not affect its mortgage lending decisions, I propose a novel irrelevance mechanism. I argue that a bank's ESG commitments are fundamentally disconnected from its actual lending outcomes. This irrelevance mechanism operates through two distinct channels: the heavy use of AUS in the mortgage market, and the design of ESG rating metrics.

The first channel is that mortgage lending decisions are, to a great extent, actually determined by AUS. When borrowers use a bank's portal to apply for a mortgage, the bank's Loan Origination System (LOS) collects standard applicant information, such as IDs, W-2s, paystubs, bank statements, tax returns, and credit reports. The LOS then automatically sends this loan file to the GSEs' AUS, which generates a definitive risk recommendation. Since banks sell about 50% to 70% of its conforming loans to GSEs like Fannie Mae and Freddie Mac through securitization (Buchak et al. (2018)), they heavily rely on these automated recommendations to make their final loan decisions. Importantly, applicant race is not used in this evaluation process. Since the actual lending decision is driven by the GSEs' automated, race-blind systems rather than the bank's own subjective preferences, a bank's corporate ESG goals do not directly impact its mortgage lending to minority groups.

The second channel is that ESG ratings do not heavily weight specific actions that support minority customers. The overall ESG rating from Refinitiv encompasses hundreds of different measures across environmental, social, and governance dimensions. When decomposing the "Social" score, only the diversity policy category contains information about a bank's social preferences regarding minority groups. However, these diversity policies predominantly focus on the internal diversity of the bank's staffs, rather than the diversity of its customer base. Although Feng et al. (2024) note that some diversity policy disclosures do include a measure for "customer diversity effort," it is only one metric among hundreds of measures, significantly downgrading its weight in the aggregate ESG rating. Furthermore, these ESG ratings are based entirely on voluntary firm disclosures, and therefore, they are subject to considerable noise. Con-

sequently, banks can easily optimize their overall ESG ratings by improving internal staff diversity and disclosures without ever fundamentally altering their customer-facing diversity.

This paper makes four primary contributions to the literature. First, it contributes to the growing literature on ESG and corporate social responsibility in the financial sector. While previous studies focus on banks' syndicated loans to polluting industries or green bond issuance, I provide micro-level evidence on how ESG ratings interact with mortgage lending. Second, I add to the extensive literature on racial disparities in the mortgage market by showing that public commitments to social responsibility do not inherently alleviate credit constraints for minority borrowers. Third, by showing how the underlying structure of the mortgage market—specifically, the rise of AUS and securitization—decouples ESG intentions from everyday local credit outcomes, I provide a concrete explanation for why current ESG ratings are largely uninformative and fail to reflect the real economic impact of financial institutions on marginalized communities. Finally, this paper exposes fundamental flaws in the construction and reliability of commercial ESG metrics. I highlight that these ratings are inherently subject to considerable noise due to their reliance on voluntary corporate disclosures, and their weighting schemes fail to accurately capture a bank's genuine inclination or efforts to support minority customers.

The findings of this paper—particularly the identification of the irrelevance mechanism—carry significant implications for policymakers, financial regulators, and ESG rating providers. Beyond the well-documented noise in ESG ratings, banks' ESG ratings are incapable of penetrating the algorithmic underwriting processes that govern mortgage loan allocation. Therefore, relying on voluntary bank ESG initiatives to alleviate racial lending disparities is fundamentally misguided.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the mortgage loan application and underwriting process. Section 4 describes the data, outlines the empirical design, and reports the baseline results. Section 5 introduces the instrumental variable approach to address endogeneity and examines the irrelevance mechanism. Section 6 provides robustness tests, and Section 7 concludes.

2 Literature Review

This paper contributes to four key strands of literature regarding mortgage lending, racial disparity, corporate social responsibility, algorithmic approval, and the potential noise of ESG metric construction.

First, this paper contributes to the growing literature examining the impact of firms' ESG practices on core business decisions. Basu et al. (2022) examine the impact of banks' ESG ratings on mortgage lending to disadvantaged communities, finding that high-ESG banks actively issue fewer mortgage loans in these localities—both in volume and dollar amount. This suggests systemic social washing, whereby banks engage in prosocial rhetoric while fundamentally reducing credit access for vulnerable communities. Similarly, Feng et al. (2024) investigate banks' diversity policies and find that such policies ironically reduce origination rates for minority borrowers, widening approval disparities. Further evidence of opportunistic ESG behavior is provided by Kim et al. (2025), who show that low-transparency borrowers readily engage in "greenwashing" to secure sustainability-linked loans (SLLs) and experience severe deterioration in their ESG performance ex post. While some studies, such as Houston and Shan (2022), find genuine propagation of ESG policies—showing that banks with high ESG ratings actively grant loans to borrowers with similar ESG profiles—this dynamic is largely observed in corporate syndicated lending rather than in mortgage lending. Unlike the existing literature, which interprets the negative relationship between ESG and minority lending as deliberate social washing in the mortgage market, this paper contributes causal evidence that a bank's corporate ESG rating does not actually have a direct, punitive impact on its mortgage lending decisions.

Second, this paper contributes to the extensive literature examining racial disparities in the residential mortgage market. A significant body of research documents the persistent existence of these disparities (Ladd (1998)). The seminal study by Munnell et al. (1996) finds that minority applicants are more than twice as likely to be denied a mortgage compared to White applicants, even after controlling for creditworthiness. Subsequent research by Bartlett et al. (2022) shows that risk-equivalent Latinx and Black borrowers pay significantly higher interest rates on GSE-securitized and FHA-insured loans, particularly in minority neighborhoods. Furthermore, Conklin et al.

(2025) find that minority homeowners are significantly more likely to face rejection on mortgage equity withdrawal (MEW) applications. In exploring the drivers of these disparities, Ambrose et al. (2021) demonstrate that the race of the broker heavily impacts the fees minority borrowers pay, while Frame et al. (2025) show that minority borrowers face higher rejection rates which only lessen when they directly engage with minority loan officers.

Conversely, some studies argue against overt differential treatment. Bhutta and Hizmo (2021) assert there are no significant racial differences in total mortgage fees, and Bhutta et al. (2025) find that denial disparities primarily stem from algorithmic approvals by race-blind AUS reacting to lower credit scores and higher leverage among minority groups, rather than differential treatment. My work bridges this literature by being one of the first to explicitly examine whether public corporate commitments—measured by banks' ESG ratings—actively intersect with or mitigate these deeply rooted racial disparities.

Third, this paper contributes to the growing literature on the implications of technological innovation and algorithmic approval in making loan decisions. Buchak et al. (2018) find that fintech lender market share in residential mortgage market nearly doubled from 2017 to 2015, and the application process provided by fintech lenders is almost entirely online and involves no human loan officer. Additionally, fintech lenders may be better able to screen potential borrowers by leveraging alternative data sources and the big data approaches inherent in technology-based lending. Bhutta et al. (2025) show that minority applicants are less likely to receive algorithmic approval from race-blind AUS, suggesting that differential treatment plays a more limited role in generating denial disparities. Complementing the existing literature, this study shows that the algorithmic approval in the mortgage loan application process means that a bank's ESG preference does not affect its lending decision in the residential mortgage market, supporting the proposed irrelevance mechanism.

Fourth, this paper contributes to the literature evaluating the reliability, noise, and potential disagreement inherent in ESG measures. Current ESG ratings are primarily assigned based on voluntary corporate disclosures, which raises serious concerns regarding their validity. Berg et al. (2022) demonstrate that available ESG measures

are highly noisy due to widespread measurement error. Houston and Shan (2022) express similar concerns, noting that mainstream ESG data rely heavily on self-reported information, which fluctuates based on opportunistic discretionary disclosures. Furthermore, Larcker et al. (2022) document significant shortcomings in the incentives of ESG rating providers, which heavily detract from the broader informativeness of their assessments. Adding to this critique, Baker et al. (2024) investigate the opportunistic use of diversity, equity, and inclusion (DEI) disclosures, documenting significant discrepancies between US-listed companies' external stances on DEI and their actual internal hiring practices, highlighting the severe consequences of selective ESG reporting. Building upon butdeparting from this existing literature, this paper expresses concerns regarding the effective coverage of the ESG ratings themselves, in addition to the noise from voluntary disclosures. I demonstrate that banks' ESG ratings systematically fail to capture a bank's true inclination to assist customers—specifically minority groups—because they fundamentally overweight internally controllable corporate policies rather than customer-facing outcomes. Acknowledging this pervasive noise and bias further confirms that real-world business outcomes should not be evaluated solely on the basis of bank-level ESG ratings.

3 The Mortgage Application and Underwriting Process

Following Bhutta et al. (2025), I outline the mortgage application and underwriting process in the US residential mortgage market to provide a clearer understanding of the irrelevance mechanism.

The vast majority of mortgage originations flow through three government-affiliated channels: (1) conventional conforming loans purchased and securitised by GSEs, primarily Fannie Mae and Freddie Mac; (2) loans insured by the Federal Housing Administration (FHA), designed for low-down-payment borrowers; and (3) loans guaranteed by the Department of Veterans Affairs (VA). Outside these programmes, depository institutions retain approximately 20% of mortgage originations on their balance sheets as portfolio loans (Bhutta et al., 2025). These portfolio loans predominantly consist of

jumbo" mortgages that exceed the conforming loan limits established annually by the Federal Housing Finance Agency (FHFA).[2]

The modern mortgage origination process is characterised by a bifurcation between data acquisition—performed by the originating bank—and quantitative credit risk assessment, conducted by the GSEs. Prospective borrowers initiate applications through commercial banks or non-bank lenders, increasingly utilising digital Loan Origination Systems (LOS) (Buchak et al., 2018; Fuster et al., 2022). During this stage, loan officers collect the requisite financial documentation, including tax returns, W-2 forms, and credit reports. Importantly, while originating institutions systematically collect applicant information and manage regulatory disclosures, their personnel do not possess unilateral authority to approve conforming loans, as lenders are bound by the automated underwriting process required to sell conforming loan to GSEs.

Following data collection, application files are transmitted via Application Programming Interfaces (APIs) to an external AUS operated by the GSEs—specifically, Fannie Mae's Desktop Underwriter (DU) or Freddie Mac's Loan Product Advisor (LPA). The AUS functions as the algorithmic gatekeeper of the conforming market, evaluating applications against proprietary statistical default models to generate a definitive recommendation regarding credit risk and programme eligibility. By federal law—specifically the Equal Credit Opportunity Act (ECOA) and Regulation B (12 CFR § 1002)—the AUS is strictly prohibited from incorporating demographic variables, including race, ethnicity, or geographic proxies such as ZIP codes.[3]

While the precise algorithms and weightings of DU and LPA remain proprietary, the GSEs publicly disclose the risk factors evaluated by their systems. These automated inputs are entirely quantitative and are categorised into credit-history metrics and non-credit structural variables. According to the *Fannie Mae Single-Family Selling Guide*,

[2]The FHFA is mandated by the Housing and Economic Recovery Act (HERA) of 2008 to set annual baseline conforming loan limits. Any mortgage exceeding these thresholds is strictly ineligible for GSE acquisition, placing it in the portfolio or private-label securitisation market.

[3]Both Fannie Mae's DU and Freddie Mac's LPA operate under the conservatorship of the FHFA. The FHFA mandates continuous fair lending reviews to ensure these automated systems comply with the ECOA, preventing algorithms from incorporating demographic data into probability-of-default calculations.

the specific factors evaluated by the AUS include:[4]

- **Credit-history risk factors:** Length of credit history, delinquent accounts, presence of active instalment loan accounts, revolving credit utilisation, public records (such as foreclosures and collection accounts), and recent credit enquiries.
- **Non-credit-history risk factors:** The borrower's equity and Loan-to-Value (LTV) ratio, Debt-to-Income (DTI) ratio, amount of verifiable liquid reserves, loan purpose (e.g., purchase versus refinance), loan term, amortisation type (fixed versus adjustable rate), occupancy type, property type, presence of co-borrowers, and cash flow assessments to verify income variability.

The AUS consolidates the evaluation of these variables into a dual-indicator recommendation: whether the default risk is sufficiently low (e.g., Accept") and whether the loan meets the algorithmic programme guidelines (e.g., Eligible"). This recommendation is mechanically binding if the originating bank intends to securitise and sell the loan to the GSEs. If a loan application does not receive an "Accept/Eligible" recommendation, the originating lender cannot unilaterally override the system without forfeiting GSE eligibility.[5] This sequenced architecture—whereby the originating bank acts primarily as a data conduit while the GSE's algorithm autonomously executes the credit decision—fundamentally eliminates the originating bank's underwriting discretion. Consequently, this institutional delegation prevents corporate social preferences from influencing the mechanical realities of the mortgage underwriting process.

[4]See the *Fannie Mae Single-Family Selling Guide*, Part B3: Underwriting Eligibility. The guide formally itemises these risk factors and states that DU evaluates credit risk "uniformly, and without regard to race, gender, or other prohibited factors."

[5]As stipulated by Freddie Mac's *Single-Family Seller/Servicer Guide* (Section 5100), lenders remain responsible for data integrity, but an AUS denial mechanically disqualifies the loan from the conventional conforming secondary market, effectively halting origination for the typical applicant.

4 Data and Empirical Design

4.1 ESG Data

I collect information on banks' ESG ratings from the Thomson Reuters/Refinitiv database, which provides ESG ratings for public companies across more than 500 different metrics since 2002. The ESG rating consists of three pillars—environmental, social, and governance— each of which consists of several subcategories with numerical scores. The environmental pillar covers resource use, emissions, and innovation; the social pillar includes workforce, human rights, community, and product responsibility; and the governance pillar encompasses management, shareholders, and CSR strategy.

Refinitiv compiles ESG information from annual reports, stock exchange filings, NGO websites, and other credible sources to produce data-driven, comparable ratings that minimise bias arising from firm size or disclosure quality. Scores for the environmental and social pillars are benchmarked against Refinitiv Business Classifications (TRBC – Industry Group), while governance scores are benchmarked against the country of incorporation.

To incorporate the effect of ESG-related controversies, Refinitiv also provides a discounted ESG score, which adjusts for negative media coverage or material controversies. To more accurately capture a firm's overall ESG performance, I use this discounted ESG rating in the analysis.

4.2 Mortgage Loan Data

Mortgage application data are drawn from the Home Mortgage Disclosure Act (HMDA), collected annually by the Federal Financial Institutions Examination Council (FFIEC). This dataset covers the vast majority of US residential mortgage applications and approvals, providing detailed information on lender identity (e.g., respondent ID, agency code), borrower characteristics (e.g., income, ethnicity, race), loan characteristics (e.g., loan amount, loan type, action taken), and property characteristics (e.g., property type and geographic location). The sample spans from 2009 to 2017.

4.3 Bank-Level Data

Bank-Level financial information comes from the Federal Reserve's Consolidated Financial Statements for Holding Companies (FR Y-9C). This dataset provides quarterly financial information for U.S. bank holding companies, including total assets, capital ratios, deposits, loan amounts, and related variables. Each institution is identified by a unique regulatory identifier (RSSD ID) assigned by the Federal Reserve System. The final sample consists of 233 bank holding companies, some of which own more than one commercial bank.

The FR Y-9C dataset identifies each bank by its RSSD ID, while the Refinitiv ESG database uses the ticker symbol to identify each firm. To merge these datasets, I proceed in several steps. First, I use the CRSP–FRB link provided by the Federal Reserve Bank of New York to connect each RSSD ID with the permanent company number (PERMCO) from the Center for Research in Security Prices (CRSP). Next, I obtain the corresponding ticker for each PERMCO using the CRSP/Compustat link from Wharton Research Data Services (WRDS). This allows me to merge the BHC financial data with Refinitiv ESG ratings through the RSSD ID–PERMCO–ticker linkage, yielding a unique set of BHC–ticker pairs. The HMDA dataset identifies lenders by respondent ID and agency code, but not by RSSD ID. To establish this link, I use the HMDA lender file provided by the Federal Reserve Bank of Philadelphia. This file matches each HMDA lender to its corresponding bank holding companies through respondent ID and RSSD ID.

Through this matching process, I identify each lender's corresponding BHC and merge the HMDA, FR Y-9C, and Refinitiv datasets. The final dataset combines BHC-level variables (ESG ratings and financial characteristics) with loan-level variables (loan amounts and application outcomes).

4.4 Summary Statistics

In this section, I describe the data sample, summarize the key variables, and present the main empirical analysis. Table 1 provides the definition for the key variables.

Table 2, Panel A reports summary statistics for bank-level variables. Over the sample period from 2009 to 2017, the mean ESG score is 0.354 with a standard deviation of

0.126. The avearage logarithm of total assets is 16.90, corresponding to approximately USD 21.86 billion. The average Tier 1 capital ratio is 13.30%, the mean capital structure ratio (total liabilities to total equity) is 8.28, and the average deposit-to-loan ratio is 1.252.

Panel B of Table 2 presents summary statistics for the loan-level variables. The average loan amount is USD 0.215 million, and the mean approval rate is 0.73, indicating that 73% of mortgage applications are approved.

Regarding borrower race, approximately 14.5% of applicants belong to minority groups, of which 6.60% are Asian, 6.73% are Black, and 1.13% are American Indian or Native Hawaiian. The remaining 85.5% are White applicants.

The mean annual applicant income is USD 0.112 million. The average population per census tract is 5,548, with a mean minority population share of 27.13%. The mean tract-to-MSA/MD income ratio is 119.1%, suggesting that the tract median family income is generally higher than the corresponding MSA/MD median income. The average HUD median family income is USD 69,135. The mean numbers of one-to-four family housing units and owner-occupied units are 1,928 and 1,502, respectively.

4.5 Empirical Approach and Results

The central premise of this study is to examine how a bank's ESG performance affects its lending behavior in the US residential mortgage market. Specifically, I test whether banks with higher ESG ratings are more likely to approve mortgage applications and whether they lend more to minority borrowers.

I estimate the following model for each bank–loan–census tract–year observation:

$$\begin{aligned} LendingOutcome_{ijct} = \alpha + \beta_1 ESG_{jt} + \beta_2 ESG_{jt} \times Minority_{it} + \beta_3 Minority_{it} \\ + \beta_4 Controls_{jt} + \beta_5 Controls_{it} + \beta_6 Controls_{ct} + \lambda_{ct} + \epsilon_{ijct} \end{aligned} \tag{1}$$

where subscripts i, j, c, and t denote the loan, bank, census tract, and year, respectively. The dependent variable *LendingOutcome* is either (1) *LoanApproval*, a dummy equal to one if the loan application is approved, or (2) *LoanAmount*, the loan amount granted to an approved applicant. *ESG* represents the bank's annual ESG rating. *Minority* represents either (a) a dummy variable for all minority applicants combined

(*Race*), or (b) separate dummies for specific racial groups (*American Indian, Asian, Black, or Native Hawaiian*).

The vector $Controls_{jt}$ includes bank-level variables: *LnAsset* is the natural logarithm of bank's total assets, *Tier1Ratio* is the Tier 1 risk-based capital ratio, *CapitalStructure* is the ratio of bank's total liabilities to total equity, and *DeposittoLoan* is the ratio of bank's total deposits to total loans.

$Controls_{it}$ is a vector of loan-level variables, which inlcudes the applicant's annual income, *ApplicantIncome*.

$Controls_{ct}$ is a vector of census tract-level variables, where *Population* is total population within a census tract, *MinorityPopulation* denotes the minority population as a percentage of the total population in a census tract, *HudMedianFamilyIncome* represents the HUD median family income, *Numberof1to4Family* denotes the number of dwellings built to house fewer than five families, *NumberofOwneroccupied* is the number of dwellings, including individual condominiums, that are owner-occupied, and *TracttoMSAMDInc* is the census tract's median family income as a percentage of the MSA/MD.

The unit of observation is the bank-loan-census tract-year. β_1 captures the effect of bank's ESG rating on the lending outcome overall, while β_2 measures how this effect differs for minority applicants. The coefficient of primary interest is β_2. I include census tract-year fixed effects, and cluster standard errors at the census tract-year level.

4.6 Empirical Results

Table 3 presents the main regression results using Equation (1) with the dependent variable being *LoanApproval*. Model (1) compares loan approvals for all minority applicants (*Race*) with those for White applicants. The coefficients on both *ESG* and $Race \times ESG$ are negative and statistically significant at the 1% level, indicating that banks with higher ESG ratings are less likely to approve mortgage applications for minority borrowers relative to White borrowers.

I further disaggregate the minority group into American Indian, Asian, Black, and Native Hawaiian applicants, and compare their loan approval rate with those of White applicants. In Model (2), the coefficient on $AmericanIndian \times ESG$ is –0.2130, sig-

nificant at the 10% level, suggesting that a one-unit increase in ESG score reduces the approval rate by 0.21 units relative to White applicants. Similarly, Model (3) – (5) report the application results for the *Asian, Black, and Native Hawaiian*, respectively. Specifically, the coefficients on $Black \times ESG$ in Model (4), and $NativeHawaiian \times ESG$ in Model (5), are equal to -0.2403, and -0.2837, and significant at the 5% level and 10% level, respectively. These findings indicate that banks with higher ESG ratings approve fewer applications from minority groups, particularly for American Indian, Black, and Native Hawaiian applicants.

Table 4 presents regression results with *LoanAmount* as the dependent variable. Model (1) compares loan amounts for all minority applicants versus White applicants. The coefficients on both ESG and $Race \times ESG$ are negative and statistically significant at the 1% level, suggesting that banks with higher ESG ratings issue smaller loan amounts to minority applicants.

Models (2)- (5) report the results for each racial group. Model (2) shows that the coefficient on $AmericanIndian \times ESG$ is -44.7543 and significant at the 1% level, indicating that a one-unit increase in ESG ratings reduces loan amounts to American Indian by approximately 44.75 units relative to White borrowers. Similarly, Models (3)–(5) report the results for *Asian*, *Black*, and *Native Hawaiian* applicants, respectively. Specifically, the coefficients on $Black \times ESG$ in Model (4) and $NativeHawaiian \times ESG$ in Model (5) are -37.85 and -30.90, and significant at the 1% and 10% levels, respectively. In contrast, in Model (3), the coefficient on $Asian \times ESG$ is positive, suggesting that banks with higher ESG ratings lend more to Asian borrowers relative to White borrowers.

Overall, the analysis indicates that banks with higher ESG ratings tend to issue fewer and smaller mortgage loans to minority applicants, with the exception of Asians, who appear to receive relatively more favorable treatment.

5 Identification Strategy

The empirical results in Section 4 reveal a negative relationship between a bank's ESG rating and its lending outcomes. However, this may be subject to common identification challenges. Low mortgage approval rates or lending amounts may reflect weak mortgage

demand from minority groups rather than banks' ESG ratings. Furthermore, reverse causality may exist: a bank's lending performance could influence its ESG rating. For example, if a bank's mortgage lending business grows slowly, it might strategically engage in more ESG activities to project a prosocial image, build reputation among customers, and attract more borrowers—essentially engaging in greenwashing. In this case, the observed negative relationship could partly reflect reverse causality rather than a true causal effect.

To address endogeneity and establish a causal link between a bank's ESG performance and lending behaviour, I employ an instrumental variable (IV) that provides plausibly exogenous variation in the bank's ESG rating. Banks typically operate numerous branches across different regions, and their overall ESG practices are influenced by local norms and policies. In particular, bank staff in branches located in Democrat-leaning counties tend to be more supportive of ESG initiatives (Di Giuli and Kostovetsky (2014), Basu et al. (2022)). Accordingly, I instrument a bank's ESG rating in a given year with the average share of votes for the Democratic presidential candidate in the most recent presidential election across all counties in which the bank's branches are located. I refer to this instrument as $DEMRATE$, defined as the total number of votes for the Democratic presidential candidate in these counties divided by the total votes cast in the same counties.

This instrument introduces time-series variation in a bank's ESG rating because county-level political preferences evolve over time. Moreover, since branch locations are largely predetermined and rarely adjusted in response to election outcomes, the instrument is plausibly exogenous. The key exclusion restriction is that county-level voting patterns affect a bank's lending decisions only through their influence on its ESG performance—a standard but untestable assumption. County-level voting data are obtained from the MIT Election Data and Science Lab and branch-level information from the Federal Deposit Insurance Corporation (FDIC).

I employ 2SLS model to replicate the bank–loan–census tract–year analysis presented in Tables 3 and 4. Because the lending outcome variables are measured at the bank–loan–census tract–year level while ESG ratings are observed at the bank–year level, the first and second stages differ in their unit of observation.

In the first stage, I regress the ESG rating on $DEMRATE$ and a set of bank-level controls. In the second stage, I regress the lending outcome variables on the fitted values of the ESG rating from the first stage and its interactions with minority indicators, along with the full set of control variables. I include census tract–year fixed effects and cluster standard errors at the census tract-year level.

Before discussing the regression results, Table 2, Panel C presents summary statistics for the 2SLS analysis. Panel C shows that, Over the 2009–2017 period, the mean value of $DEMRATE$ is 55.1%, suggesting that the majority of votes were cast for Democratic presidential candidates. For the bank-level variables, the mean values of $Tier1Ratio$, $CapitalStructure$, $LnAsset$, and $DeposittoLoan$ are 13.23%, 8.29, 16.89, and 1.24, respectively.

Tables 5 and 6 report the 2SLS results. Column (1) in both tables presents first-stage regression results with banks' ESG rating as the dependent variable to check instrument relevance. The main variable of interest is the instrument $DEMRATE$. The coefficient on $DEMRATE$ is positive and significant at the 1% level. The significantly positive coefficient on $DEMRATE$ indicates that banks with branches in Democrat-leaning areas tend to have higher ESG ratings than those in Republican-leaning areas. The strong statistical significance of the instrument confirms that the instrument is highly correlated with banks' ESG ratings. Therefore, coefficient estimates and their corresponding standard errors reported in the second stage are likely to be unbiased and inferences drawn from them are reasonably valid.

Columns (2)–(6) of Table 5 report results from the second-stage regression using Equation (1) with the main independent variable replaced by the fitted value of ESG from the first-stage regression and the dependent variable $LoanApproval$. Column (2) compares loan approvals for all minority applicants ($Race$) with those for White applicants. Columns (3) - (6) compare loan approvals for each racial group with those for White applicants. The coefficients on ESG in Columns (2) - (6) are negative but insignificant, while the coefficients on $Race \times ESG$, $AmericanIndian \times ESG$, and $NativeHawaiian \times ESG$ are positive and insignificant. The 2SLS analysis shows that banks' high ESG ratings have no impact on their loan approval rates to minority groups, supporting the third hypothesis.

Columns (2)–(6) of Table 6 report second-stage results from Equation (1), with the main independent variable replaced by the fitted value of ESG from the first stage and *LoanAmount* as the dependent variable. Column (2) compares loan amounts for all minority applicants (*Race*) with those for White applicants. Columns (3) - (6) compare loan amounts for each racial group with those for White applicants. The coefficients on ESG in Columns (2) - (6) are negative and significant at the 5% levels, while the coefficients on $AmericanIndian \times ESG$, $Asian \times ESG$, $Black \times ESG$, and $NativeHawaiian \times ESG$ are positive and insignificant. In contrast to the OLS findings, the 2SLS analysis shows that banks' ESG ratings have no causal impact on loan amounts to minority groups, further supporting the third hypothesis.

Taken together, the 2SLS results do not corroborate the baseline OLS findings, suggesting that banks' ESG ratings do not causally affect their lending decisions, supporting the irrelevance hypothesis.

My empirical findings differ from those of Basu et al. (2022) and Feng et al. (2024). Basu et al. (2022) examine how banks' mortgage lending decisions in low-income localities are affected by their ESG ratings, instrumenting for a bank's ESG rating using the Democratic presidential candidate's vote share in the state where the bank's headquarters are located in the most recent presidential election. By contrast, my paper examines how banks' mortgage lending decisions toward minority borrowers are affected by their ESG ratings, instrumenting for ESG ratings using the Democratic presidential candidate's vote share in the counties where bank branches are located. Most closely related to my study, Feng et al. (2024) focus on the role of diversity policies rather than ESG ratings. Importantly, such diversity policies are not primarily customer-facing and are unlikely to directly affect lending outcomes for minority borrowers.

6 Alternative Sample for Robustness Tests

Following an update to the HMDA data in 2018, the dataset includes more granular information about applicant and loan characteristics, such as interest rates and the use of AUS. This richer dataset enables me to test the robustness of the primary results. Therefore, in this section, I analyze the relationship between a bank's ESG performance

and its lending decisions toward minority applicants using data from 2018 to 2019, following the methodology of Bhutta et al. (2025). I examine three critical stages of the mortgage lending process: (1) loan approval, (2) loan amount, and (3) the interest rate paid by the mortgage loan borrowers. Across all specifications, I include the AUS indicator as a control variable and control for a comprehensive set of fixed effects—including County-Year, Co-Applicant, Income Bins, Loan-to-Value (LTV) Bins, and Bank Fixed Effects—while clustering standard errors at the Bank and County levels.

Table 2, Panel D reports summary statistics for the key variables. Over the sample period from 2018 to 2019, the mean ESG score is 0.481 with a standard deviation of 0.152, and the mean S score is 0.645 with a standard deviation of 0.233.

Turning to the loan-level variables, the average loan amount is USD 0.123 million, and the mean approval rate is 0.82, indicating that 82% of mortgage applications are approved. The mean interest rate is 4.3%, while the mean value of AUS is 0.71, suggesting that 71% of loan applications use AUS.

Regarding borrower race, approximately 17.5% of applicants belong to minority groups, of which 10.4% are Asian, 7.5% are Black, and 1.13% are American Indian or Native Hawaiian. The remaining 82.5% are White applicants. Compared with the 14.5% of minority share observed over the 2009–2017 period, there is an increase in the number of minority applicants.

6.1 Loan Approval and ESG Rating

In this section, using the updated HMDA data, I examine whether banks with high ESG ratings are more or less likely to approve minority loan applications. The primary variable of interest is the interaction term between the minority applicant race indicator and the bank's ESG rating.

Table 7 presents regression results with *LoanApproval* as the dependent variable. Model (1) compares loan approvals for all minority applicants (*Race*) with those for White applicants. The coefficient on $Race \times ESG$ is negative and statistically significant at the 10% level, indicating that banks with higher ESG ratings are less likely to approve mortgage applications from minority borrowers relative to White borrowers overall. However, when I further disaggregate the minority group into American Indian,

Asian, Black, and Native Hawaiian applicants to compare their specific loan approval rates with those of White applicants, the regression results are highly inconsistent. The coefficients on $AmericanIndian \times ESG$ in Model (2) and $Black \times ESG$ in Model (4) are negative and significant at the 10% and 5% levels, respectively, while the coefficients on $Asian \times ESG$ in Model (3) and $NativeHawaiian \times ESG$ in Model (5) are statistically insignificant. This inconsistency across subpopulations suggests that a bank's ESG rating does not have a reliable or systemic impact on loan approval rates.

The coefficient on AUS is significantly positive across all models, suggesting a significant progress in fair lending through the adoption of AUS. This result is consistent with Bhutta et al. (2025), who show that algorithmic approvals promote fair lending in the residential mortgage market.

6.2 Loan Amount and ESG Rating

Table 8 presents regression results with *LoanAmount* as the dependent variable. Model (1) compares the loan amounts for all minority applicants (*Race*) with those for White applicants. The coefficient on $Race \times ESG$ is negative but insignificant, indicating that a bank's ESG rating has no impact on the aggregate lending amount extended to minority borrowers. When examining specific demographics, the coefficients on $AmericanIndian \times ESG$ in Model (2), $Black \times ESG$ in Model (4), and $NativeHawaiian \times ESG$ in Model (5) are all insignificant. This lack of significance further supports my hypothesis that a bank's ESG rating has no meaningful impact on lending decisions for these minority groups. While the coefficient on $Asian \times ESG$ in Model (3) is positive and significant at the 1% level, this finding is attributable to the relatively higher average income levels typical of Asian applicants rather than a systemic ESG-driven initiative.

6.3 Interest Rate and ESG Rating

The updated HMDA data provides detailed information ont the interest rates paid by borrowers, which reflects the borrowing cost faced by applicants. Therefore, in this section, I examine the impact of banks' ESG ratings on the interest rates.

Table 9 presents regression results with *InterestRate* as the dependent variable.

Model (1) compares the interest rates offered to all minority applicants (*Race*) with those offered to White applicants. The coefficient on $Race \times ESG$ is negative and statistically significant at the 5% level. Furthermore, when analyzing specific demographic groups, the coefficients on $AmericanIndian \times ESG$ in Model (2), $Black \times ESG$ in Model (4), and $NativeHawaiian \times ESG$ in Model (5) are all statistically insignificant. Although the coefficient on $Asian \times ESG$ in Model (3) is negative and marginally significant at the 10% level, the overall lack of consistent significance across the most historically marginalized groups further supports my hypothesis that a bank's ESG rating has no systemic or economically meaningful impact on the cost of credit for minority borrowers.

6.4 Lending Decisions and the 'S' Component of ESG

Banks' mortgage lending to minority groups contributes primarily to the social aspect of bank ESG. To sharpen the identification, I isolate the "S" component of the ESG rating and replicate the main regressions from Sections 6.1, 6.2, and 6.3 by replacing the composite ESG rating with the S rating alone. The regression results are presented in Tables 10, 11, and 12. These results confirm the baseline findings and provide further evidence that a bank's ESG rating has no substantive impact on its lending decisions toward minority groups in the residential mortgage market.

7 Conclusion

This paper provides evidence on how banks' ESG ratings affect their lending decisions toward minority groups. Using a comprehensive dataset covering all US residential mortgage loans from 2009 to 2017, combined with banks' ESG ratings and an instrumental variable identification strategy, I identify a key mechanism linking ESG ratings to mortgage lending decisions.

I find that banks' ESG ratings have no impact on their lending decisions toward minority groups, due to what I term the irrelevance mechanism. Specifically, mortgage lending decisions are largely determined by AUS provided by GSEs, which do not incorporate banks' ESG preferences. Furthermore, ESG metrics tend to prioritize internal

workforce diversity over customer-facing diversity.

A series of robustness checks confirms that these results are consistent across different samples and empirical methods. Overall, this study offers new insights into the real effects of banks' ESG ratings on their lending decisions toward minority groups in the residential mortgage market.

The findings of this paper—particularly the identification of the irrelevance mechanism—carry significant implications for policymakers, financial regulators, and ESG rating providers. Beyond the well-documented noise in ESG ratings, corporate-level ESG ratings are unable to penetrate the algorithmic underwriting processes that govern mortgage loan allocation. Therefore, relying on voluntary bank ESG initiatives to alleviate racial lending disparities is fundamentally misguided.

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Table 1 Variable Definitions

Variable	Definition
LoanApproval	LoanApproval is a dummy variable. It is equal to one if the mortgage loan application is approved; otherwise, it is zero.
LoanAmount	LoanAmount is the loan amount the lender issue to each successful mortgage applicant.
Race	Race refers to the minority group.
DEMRATE	DEMRATE is calculated by summing the votes for the Democratic presidential candidate across all counties where bank's branches are located divided by the sum of total votes in these counties.
ESG	ESG represents bank's annual ESG score, which is come from Refinitive.
S	ESG represents bank's annual S score, which is come from Refinitive.
AmericanIndian	AmericanIndian is equal to one if the applicant is the American Indian or Alaska Native.
Asian	Asian equals one if the applicant is Asian.
AUS	Automatic underwriting system.
Black	Black equals one if the applicant is Black.
NativeHawaiian	This value is equal to one if the applicant is Native Hawaiian or Other Pacific Islander
ApplicantIncome	The annual income of applicant
MinorityPopulation	Percentage of minority population to total population for tract
Population	Total population in tract
MedFamilyInc	FFIEC Median Family Income: FFIEC Median family income in dollars for the MSA/MD in which the tract is located (adjusted annually by FFIEC).
TracttoMSAMDInc	Tract to MSA/MD Median Family Income Percentage: % of tract median family income compared to MSA/MD median family income. (Carried to two decimal places)
NumofOwnerOccupied	Number of Owner Occupied Units: Number of dwellings, including individual condominiums, that are lived in by the owner.
Numof1-4Family	Number of 1- to 4-Family units: Dwellings that are built to house fewer than 5 families.
LnAsset	The logarithm of bank's total asset.
Tier1Ratio	It is risk-based capital ratio
CapitalStructure	It is the ratio of total liability to total equity.
DeposittoLoan	It is the ratio of total deposit to total loan.

Table 2 Summary Statistics

Panel A Summary statistics for variables at the bank-level. This table summarizes the sample statistics of key bank-level characteristics used in our study. Panel A uses the sample of banks during the 2009-2017 period.

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
ESG	689	0.354	0.126	0.205	0.273	0.346	0.419	0.510	0.0230	0.815
Tier1Ratio	669	13.30	5.013	10.37	11.26	12.42	14.12	16.50	8.441	97.16
CapitalStructure	675	8.277	1.915	6.229	6.948	8.130	9.304	10.73	0.278	15.38
LnAsset	675	16.90	1.928	14.56	15.45	16.71	18.04	19.70	13.66	21.67
DeposittoLoan	675	1.252	0.559	0.939	1.016	1.106	1.273	1.598	0.593	5.276

Panel B: Summary statistics for variables at the loan-level. Panel B summarizes the sample statistics of key loan-level characteristics used in our study. Panel B uses the sample of mortgage loan application and origination during the 2009-2017 period.

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
LoanAmount	4.024e+07	215.3	221.5	60	103	167	266	403	1	99,999
ApplicantIncome	3.738e+07	111.8	174.3	33	50	79	125	200	1	44,560
Population	3.985e+07	5,548	2,791	2,815	3,794	5,085	6,671	8,484	0	53,812
MinorityPopulation	3.984e+07	27.13	24.75	4.290	8.540	18.35	37.61	67.14	0	100
MedFamilyInc	3.985e+07	69,135	14,598	52,800	59,900	67,200	75,900	88,800	14,400	131,500
TracttoMSAMDInc	3.984e+07	119.1	42.38	75	92.13	112.2	138	171.0	0	507.5
NumofOwnerOccupied	3.983e+07	1,502	811.4	665	978	1,381	1,861	2,415	0	19,529
Numof1-4Family	3.984e+07	1,928	999.9	932	1,299	1,766	2,352	3,050	0	25,391
Tier1Ratio	4.018e+07	11.90	1.443	10.42	11.10	11.81	12.54	13.68	8.441	97.16
CapitalStructure	4.023e+07	8.660	1.248	7.039	8.023	8.415	8.981	10.56	0.278	15.38
LnAsset	4.023e+07	20.53	1.427	18.42	19.81	21.08	21.47	21.54	13.66	21.67
DeposittoLoan	4.023e+07	1.202	0.211	1.000	1.046	1.157	1.313	1.508	0.593	5.276
Race	4.024e+07	0.145	0.352	0	0	0	0	1	0	1
AmericanIndian	4.024e+07	0.00676	0.0819	0	0	0	0	0	0	1
Asian	4.024e+07	0.0660	0.248	0	0	0	0	0	0	1
Black	4.024e+07	0.0673	0.250	0	0	0	0	0	0	1
NativeHawaiian	4.024e+07	0.00451	0.0670	0	0	0	0	0	0	1
White	4.024e+07	0.855	0.352	0	1	1	1	1	0	1
LoanApproval	4.024e+07	0.729	0.444	0	0	1	1	1	0	1

Panel C: Summary statistics for variables used in the instrumental variable analysis. Panel C summarizes the sample statistics of key variables used in my study, and it uses the sample of the vote for the presidential candidate during the 2009-2017 period.

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
ESG	665	0.356	0.127	0.205	0.276	0.348	0.420	0.511	0.0230	0.815
Tier1Ratio	652	13.23	5.002	10.38	11.26	12.40	14.05	16.33	8.441	97.16
CapitalStructure	652	8.291	1.876	6.301	7.002	8.155	9.285	10.61	0.278	15.38
LnAsset	652	16.89	1.935	14.57	15.45	16.68	18.00	19.70	13.66	21.67
DeposittoLoan	652	1.241	0.536	0.939	1.017	1.106	1.265	1.592	0.593	5.276
DEMRATE	665	0.551	0.0993	0.423	0.481	0.552	0.620	0.684	0.252	0.787

Panel D: Summary statistics for variables used in robustness tests. Panel D presents summary statistics for the key variables used in the robustness tests, covering the 2018–2019 period.

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
ESG	2.503e+06	0.481	0.152	0.293	0.418	0.457	0.523	0.739	0.0655	0.814
S	2.503e+06	0.645	0.233	0.273	0.446	0.723	0.837	0.869	0.0091	0.920
LoanApproval	2.503e+06	0.823	0.382	0	1	1	1	1	0	1
AmericanIndian	2.084e+06	0.00880	0.0934	0	0	0	0	0	0	1
Asian	2.307e+06	0.104	0.306	0	0	0	0	1	0	1
Black	2.234e+06	0.0755	0.264	0	0	0	0	0	0	1
NativeHawaiian	2.075e+06	0.00448	0.0668	0	0	0	0	0	0	1
Race	2.503e+06	0.175	0.380	0	0	0	0	1	0	1
LoanAmount	2.503e+06	12.31	0.797	11.35	11.81	12.28	12.83	13.36	10.13	14.38
InterestAmount	2.038e+06	0.0426	0.212	0.0338	0.0375	0.0425	0.0463	0.0501	0	302.5
AUS	2.067e+06	0.710	0.454	0	0	1	1	1	0	1

Table 3 Banks' ESG ratings and the origination of mortgage loans. This table presents the results from the models for our preliminary analysis used in Section 4. The dependent variables are the origination of mortgage loans. The key explanatory variables are ESG ratings and racial groups. All models use annual data for 2009-2017. T-statistics reported in parentheses are based on robust standard errors double-clustered by the census tract and by year. All models include census tract x year fixed effects. Superscripts ∗∗∗, ∗∗, and ∗ indicate statistical significance at 1%, 5%, and 10% levels.

VARIABLES	(1) LoanApproval	(2) LoanApproval	(3) LoanApproval	(4) LoanApproval	(5) LoanApproval
ESG	-0.5200***	-0.5172***	-0.5179***	-0.5195***	-0.5173***
	(0.0828)	(0.0821)	(0.0827)	(0.0821)	(0.0820)
Race x ESG	-0.1756***				
	(0.0440)				
AmericanIndian x ESG		-0.2130*			
		(0.1193)			
Asian x ESG			-0.0444		
			(0.0939)		
Black x ESG				-0.2403**	
				(0.1068)	
NativeHawaiian x ESG					-0.2837***
					(0.0826)
Race	-0.0019				
	(0.0198)				
AmericanIndian		-0.0559			
		(0.0519)			
Asian			-0.0140		
			(0.0393)		
Black				-0.0088	
				(0.0544)	
NativeHawaiian					0.0272
					(0.0352)
ApplicantIncome	0.0000	0.0000	0.0000	0.0000	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
MinorityPopulation	-0.0014***	-0.0012***	-0.0011***	-0.0013***	-0.0012***

	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Population	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
MedFamilyInc	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
TracttoMSAMDInc	0.0004***	0.0003***	0.0003***	0.0003***	0.0003***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
NumofOwnerOccupied	-0.0000	-0.0000	0.0000	-0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Numof1-4Family	-0.0000***	-0.0000**	-0.0000**	-0.0000***	-0.0000**
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
LnAsset	0.0208***	0.0205***	0.0196***	0.0216***	0.0204***
	(0.0053)	(0.0054)	(0.0055)	(0.0052)	(0.0054)
Tier1Ratio	-0.0165	-0.0160	-0.0166	-0.0161	-0.0160
	(0.0129)	(0.0126)	(0.0126)	(0.0129)	(0.0126)
CapitalStructure	-0.0114	-0.0123	-0.0116	-0.0124	-0.0123
	(0.0109)	(0.0105)	(0.0105)	(0.0109)	(0.0105)
DeposittoLoan	0.0829	0.0900	0.0884	0.0857	0.0900
	(0.0626)	(0.0594)	(0.0609)	(0.0607)	(0.0591)
Constant	0.5985**	0.6056**	0.6276***	0.5921**	0.6064**
	(0.2417)	(0.2387)	(0.2364)	(0.2431)	(0.2388)
Observations	36,919,538	31,920,690	34,151,048	34,029,756	31,837,908
R-squared	0.0458	0.0394	0.0372	0.0470	0.0383
Census tract-year FE	Yes	Yes	Yes	Yes	Yes

Table 4 Banks' ESG ratings and the amount of mortgage loans. This table presents the results from the models for our preliminary analysis used in Section 4. The dependent variables are the amount of mortgage loans. The key explanatory variables are ESG ratings and racial groups. All models use annual data for 2009-2017. T-statistics reported in parentheses are based on robust standard errors double-clustered by the census tract and by year. All models include census tract x year fixed effects. Superscripts ∗∗∗, ∗∗, and ∗ indicate statistical significance at 1%, 5%, and 10% levels.

VARIABLES	(1) LoanAmount	(2) LoanAmount	(3) LoanAmount	(4) LoanAmount	(5) LoanAmount
ESG	-59.8323***	-57.7578***	-59.4245***	-58.2369***	-57.7665***
	(21.5152)	(22.1398)	(22.3726)	(21.6413)	(22.1844)
Race x ESG	-13.4928*				
	(7.2342)				
AmericanIndian x ESG		-44.7543***			
		(14.0748)			
Asian x ESG			40.7411***		
			(12.5796)		
Black x ESG				-37.8530***	
				(13.8132)	
NativeHawaiian x ESG					-30.9002*
					(18.6891)
Race	5.7667*				
	(3.2589)				
AmericanIndian		-4.6102			
		(8.3115)			
Asian			2.4333		
			(5.4857)		
Black				-1.5688	
				(6.3739)	
NativeHawaiian					11.8020
					(7.9578)
ApplicantIncome	0.4644***	0.4659***	0.4686***	0.4625***	0.4667***
	(0.0261)	(0.0263)	(0.0260)	(0.0262)	(0.0263)
MinorityPopulation	0.4169***	0.5630***	0.5924***	0.4371***	0.5706***

	(0.0339)	(0.0363)	(0.0343)	(0.0346)	(0.0355)
Population	0.0064***	0.0048***	0.0048***	0.0059***	0.0048***
	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0004)
MedFamilyInc	0.0036***	0.0036***	0.0037***	0.0035***	0.0036***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
TracttoMSAMDInc	1.6863***	1.6575***	1.6692***	1.6612***	1.6580***
	(0.0726)	(0.0762)	(0.0742)	(0.0740)	(0.0759)
NumofOwnerOccupied	-0.0633***	-0.0590***	-0.0590***	-0.0616***	-0.0589***
	(0.0027)	(0.0026)	(0.0026)	(0.0026)	(0.0026)
Numof1-4Family	0.0264***	0.0277***	0.0274***	0.0270***	0.0277***
	(0.0019)	(0.0020)	(0.0020)	(0.0019)	(0.0020)
LnAsset	4.6246***	4.4714***	4.5287***	4.5776***	4.4561***
	(1.2575)	(1.2613)	(1.2630)	(1.2483)	(1.2651)
Tier1Ratio	2.5206	2.3571	2.2942	2.4325	2.3368
	(3.0121)	(3.1691)	(3.1024)	(3.0992)	(3.1750)
CapitalStructure	3.5124	3.1006	2.9318	3.4315	3.0791
	(3.5915)	(3.7261)	(3.6791)	(3.6341)	(3.7316)
DeposittoLoan	-9.8020	-7.8791	-6.7589	-9.6142	-7.6044
	(26.5819)	(28.2761)	(27.3827)	(27.6431)	(28.3496)
Constant	-409.6450***	-404.7386***	-408.1177***	-401.1282***	-404.9234***
	(53.1453)	(55.0337)	(54.3248)	(53.9902)	(55.1016)
Observations	36,919,538	31,920,690	34,151,048	34,029,756	31,837,908
R-squared	0.4659	0.4575	0.4709	0.4538	0.4578
Census tract-year FE	Yes	Yes	Yes	Yes	Yes

Table 5 2SLS instrumental variable approach. This table presents the results from replicating the models for our preliminary analysis used in Section 5. The dependent variables are the origination of mortgage loans. The key explanatory variables are predicted banks' ESG ratings and racial groups. All models use annual data for 2009-2017. Column (1) reports the first-stage regression result. T-statistics reported in parentheses are based on robust standard errors double-clustered by the census tract and by year. All models include census tract x year fixed effects. Superscripts ∗∗∗, ∗∗, and ∗ indicate statistical significance at 1%, 5%, and 10% levels.

VARIABLES	(1) ESG	(2) LoanApproval	(3) LoanApproval	(4) LoanApproval	(5) LoanApproval	(6) LoanApproval
esghat		-395.6734	-449.0080	-428.3967	-448.5357	-448.6070
		(461.5185)	(459.6467)	(456.4889)	(460.9263)	(460.8251)
DEMRATE	0.1597***					
	(0.0438)					
Race x ESG		63.7986*				
		(35.1723)				
AmericanIndian x ESG			26.8686			
			(64.7277)			
Asian x ESG				19.5846		
				(37.8510)		
Black x ESG					54.8186	
					(41.5072)	
NativeHawaiian x ESG						4.0197
						(73.1944)
Race		-29.1966*				
		(16.6756)				
AmericanIndian			-35.2648			
			(27.4377)			
Asian				9.8469		
				(18.0366)		
Black					-42.6124**	
					(19.3110)	

NativeHawaiian						-2.4533
						(34.4466)
ApplicantIncome		0.4639***	0.4655***	0.4681***	0.4620***	0.4663***
		(0.0263)	(0.0265)	(0.0262)	(0.0264)	(0.0265)
MinorityPopulation		0.4391***	0.5891***	0.6182***	0.4603***	0.5969***
		(0.0397)	(0.0450)	(0.0424)	(0.0423)	(0.0442)
Population		0.0064***	0.0047***	0.0047***	0.0059***	0.0047***
		(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0004)
MedFamilyInc		0.0036***	0.0036***	0.0037***	0.0035***	0.0036***
		(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
TracttoMSAMDInc		1.6881***	1.6585***	1.6704***	1.6627***	1.6590***
		(0.0729)	(0.0761)	(0.0743)	(0.0742)	(0.0759)
NumofOwnerOccupied		-0.0631***	-0.0586***	-0.0586***	-0.0613***	-0.0585***
		(0.0025)	(0.0023)	(0.0024)	(0.0024)	(0.0023)
Numof1-4Family		0.0265***	0.0278***	0.0275***	0.0270***	0.0278***
		(0.0019)	(0.0020)	(0.0020)	(0.0019)	(0.0020)
LnAsset	0.0265***	15.4227	17.0658	16.4807	17.0807	17.0487
	(0.0025)	(13.2169)	(13.1830)	(13.0697)	(13.2246)	(13.2150)
Tier1Ratio	-0.0011*	3.5677	3.2922	3.2443	3.4005	3.2702
	(0.0007)	(2.9238)	(3.0442)	(2.9778)	(2.9875)	(3.0491)
CapitalStructure	-0.0059***	2.0169	1.2161	1.1673	1.6148	1.1995
	(0.0023)	(3.6164)	(3.5579)	(3.5418)	(3.5709)	(3.5704)
DeposittoLoan	0.0249***	-3.6115	0.0545	0.5975	-1.9279	0.2545
	(0.0093)	(25.1783)	(26.5128)	(25.7410)	(26.0418)	(26.6150)
Constant	-0.1453***	-480.9440***	-484.5388***	-485.5949***	-480.0808***	-484.8348***
	(0.0406)	(74.1891)	(76.8205)	(76.1531)	(75.2743)	(76.8118)
Observations	652	36,749,566	31,769,215	33,995,169	33,866,664	31,686,720
R-squared	0.2571	0.4655	0.4571	0.4706	0.4533	0.4574
Census tract-year FE	No	Yes	Yes	Yes	Yes	Yes

Table 6 2SLS instrumental variable approach. This table presents the results from replicating the models for our preliminary analysis used in Section 5. The dependent variables are the amount of mortgage loans. The key explanatory variables are predicted banks' ESG ratings and racial and ethnical group. All models use annual data for 2009-2017. Column (1) reports the first-stage regression result. T-statistics reported in parentheses are based on robust standard errors double-clustered by the census tract and by year. All models include census tract x year fixed effects. Superscripts ∗∗∗, ∗∗, and ∗ indicate statistical significance at 1%, 5%, and 10% levels.

VARIABLES	(1) ESG	(2) LoanAmount	(3) LoanAmount	(4) LoanAmount	(5) LoanAmount	(6) LoanAmount
ESG		-5.3415**	-5.4767**	-5.4262**	-5.5442**	-5.4586**
		(2.3685)	(2.3078)	(2.2918)	(2.3518)	(2.3092)
DEMRATE	0.1597***					
	(0.0438)					
Race x ESG		0.1026				
		(0.2509)				
AmericanIndian x ESG			0.4408			
			(0.3310)			
Asian x ESG				-0.6817**		
				(0.2992)		
Black x ESG					0.6852**	
					(0.3292)	
NativeHawaiian x ESG						0.2266
						(0.3978)
Race		-0.1195				
		(0.1181)				
AmericanIndian			-0.3477**			
			(0.1467)			
Asian				0.2908**		
				(0.1431)		
Black					-0.4270***	
					(0.1414)	

NativeHawaiian						-0.1922
						(0.1868)
ApplicantIncome		0.0000	0.0000	0.0000	0.0000	0.0000
		(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
MinorityPopulation		-0.0012***	-0.0010***	-0.0009***	-0.0011***	-0.0010***
		(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Population		0.0000***	0.0000***	0.0000***	0.0000***	0.0000***
		(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
MedFamilyInc		0.0000***	0.0000***	0.0000***	0.0000***	0.0000***
		(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
TracttoMSAMDInc		0.0004***	0.0004***	0.0004***	0.0004***	0.0004***
		(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
NumofOwnerOccupied		-0.0000	0.0000	0.0000	-0.0000	0.0000
		(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Numof1-4Family		-0.0000***	-0.0000**	-0.0000**	-0.0000***	-0.0000**
		(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
LnAsset	0.0265***	0.1702**	0.1746***	0.1729***	0.1766***	0.1740***
	(0.0025)	(0.0678)	(0.0662)	(0.0657)	(0.0674)	(0.0662)
Tier1Ratio	-0.0011*	-0.0059	-0.0068	-0.0070	-0.0066	-0.0067
	(0.0007)	(0.0127)	(0.0122)	(0.0122)	(0.0124)	(0.0122)
CapitalStructure	-0.0059***	-0.0332	-0.0355	-0.0350	-0.0354	-0.0353
	(0.0023)	(0.0224)	(0.0217)	(0.0215)	(0.0222)	(0.0217)
DeposittoLoan	0.0249***	0.1715	0.1869*	0.1842*	0.1821*	0.1862*
	(0.0093)	(0.1072)	(0.1006)	(0.1027)	(0.1026)	(0.1006)
Constant	-0.1453***	-0.2725	-0.2769	-0.2673	-0.2837	-0.2759
	(0.0406)	(0.2613)	(0.2546)	(0.2557)	(0.2577)	(0.2545)
Observations	652	36,749,566	31,769,215	33,995,169	33,866,664	31,686,720
R-squared	0.2571	0.0399	0.0347	0.0326	0.0417	0.0336
Census tract-year FE		Yes	Yes	Yes	Yes	Yes

Table 7 Banks' ESG ratings and the origination of mortgage loans. This table presents the regression results in Section 6.1. The dependent variables are the origination of mortgage loans. The key explanatory variables are ESG ratings and racial groups. All models use annual data for 2018-2019. T-statistics reported in parentheses are based on robust standard errors double-clustered by the bank and by year. All models include County-Year, Co-Applicant, Income Bins, Loan-to-Value (LTV) Bins, and Bank fixed effects. Superscripts ∗∗∗, ∗∗, and ∗ indicate statistical significance at 1%, 5%, and 10% levels.

VARIABLES	(1) LaonApproval	(2) LaonApproval	(3) LaonApproval	(4) LaonApproval	(5) LaonApproval
ESG	-0.0938	-0.0858	-0.0870	-0.0941	-0.0895
	(0.0707)	(0.0681)	(0.0652)	(0.0725)	(0.0675)
Race x ESG	-0.0388*				
	(0.0198)				
AmericanIndian x ESG		-0.0482*			
		(0.0283)			
Asian x ESG			-0.0122		
			(0.0199)		
Black x ESG				-0.0554**	
				(0.0215)	
NativeHawaiian x ESG					-0.00435
					(0.0547)
Race	-0.0271***				
	(0.00917)				
AmericanIndian		-0.0259			
		(0.0170)			
Asian			-0.0225**		
			(0.00894)		
Black				-0.0377***	
				(0.0113)	
NativeHawaiian					-0.0716*
					(0.0411)
AUS	0.254***	0.240***	0.240***	0.249***	0.238***
	(0.0510)	(0.0478)	(0.0475)	(0.0507)	(0.0472)
Constant	0.725***	0.735***	0.737***	0.728***	0.738***
	(0.0433)	(0.0414)	(0.0402)	(0.0438)	(0.0410)

Observations	1,811,345	1,532,091	1,667,023	1,641,769	1,525,720
R-squared	0.303	0.297	0.293	0.305	0.297
County-Year	Yes	Yes	Yes	Yes	Yes
Co-Applicant FE	Yes	Yes	Yes	Yes	Yes
Income Bins	Yes	Yes	Yes	Yes	Yes
LTV Bins	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes

Table 8 Banks' ESG ratings and the amount of mortgage loans. This table presents the regression results in Section 6.2. The dependent variables are the amount of mortgage loans. The key explanatory variables are ESG ratings and racial groups. All models use annual data for 2018-2019. T-statistics reported in parentheses are based on robust standard errors double-clustered by the bank and by year. All models include County-Year, Co-Applicant, Income Bins, Loan-to-Value (LTV) Bins, and Bank fixed effects. Superscripts ∗∗∗, ∗∗, and ∗ indicate statistical significance at 1%, 5%, and 10% levels.

VARIABLES	(1) LoanAmount	(2) LoanAmount	(3) LoanAmount	(4) LoanAmount	(5) LoanAmount
ESG	0.00187	0.0194	0.00286	0.0169	0.0174
	(0.0349)	(0.0311)	(0.0319)	(0.0342)	(0.0311)
Race x ESG	-0.0196				
	(0.0269)				
AmericanIndian x ESG		-0.000514			
		(0.0465)			
Asian x ESG			0.102***		
			(0.0332)		
Black x ESG				-0.0826	
				(0.0707)	
NativeHawaiian x ESG					0.0330
					(0.0558)
Race	-0.00620				
	(0.0123)				
AmericanIndian		-0.0552**			
		(0.0217)			
Asian			0.0106		
			(0.0162)		
Black				-0.0536*	
				(0.0320)	
NativeHawaiian					-0.0584
					(0.0380)
AUS	0.0570***	0.0550***	0.0491***	0.0601***	0.0546***
	(0.0110)	(0.0106)	(0.0108)	(0.0105)	(0.0106)
Constant	12.17***	12.13***	12.18***	12.13***	12.14***
	(0.0198)	(0.0181)	(0.0187)	(0.0194)	(0.0180)

Observations	1,811,345	1,532,091	1,667,023	1,641,769	1,525,720
R-squared	0.790	0.789	0.795	0.786	0.789
County-Year	Yes	Yes	Yes	Yes	Yes
Co-Applicant FE	Yes	Yes	Yes	Yes	Yes
Income Bins	Yes	Yes	Yes	Yes	Yes
LTV Bins	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes

Table 9 Banks' ESG ratings and interest rate. This table presents the regression results in Section 6.3. The dependent variables are the mortgage loans' interest rates. The key explanatory variables are ESG ratings and racial groups. All models use annual data for 2018-2019. T-statistics reported in parentheses are based on robust standard errors double-clustered by the bank and by year. All models include County-Year, Co-Applicant, Income Bins, Loan-to-Value (LTV) Bins, and Bank fixed effects. Superscripts ∗∗∗, ∗∗, and ∗ indicate statistical significance at 1%, 5%, and 10% levels.

VARIABLES	(1) InterestRate	(2) InterestRate	(3) InterestRate	(4) InterestRate	(5) InterestRate
ESG	-0.00400	-0.00524	-0.00417	-0.00474	-0.00507
	(0.00588)	(0.00637)	(0.00618)	(0.00612)	(0.00639)
Race x ESG	-0.00131**				
	(0.000635)				
AmericanIndian x ESG		-0.00347			
		(0.00281)			
Asian x ESG			-0.00177*		
			(0.00104)		
Black x ESG				-0.000849	
				(0.00111)	
NativeHawaiian x ESG					0.000509
					(0.00120)
Race	0.000324				
	(0.000285)				
AmericanIndian		0.00192			
		(0.00145)			
Asian			-0.000225		
			(0.000449)		
Black				0.000974**	
				(0.000383)	
NativeHawaiian					0.000137
					(0.000749)
AUS	-0.00309***	-0.00325***	-0.00301***	-0.00320***	-0.00320***
	(0.000937)	(0.000948)	(0.000923)	(0.000953)	(0.000942)
Constant	0.0476***	0.0484***	0.0476***	0.0482***	0.0483***
	(0.00271)	(0.00297)	(0.00286)	(0.00285)	(0.00298)

Observations	1,543,238	1,319,979	1,436,815	1,400,703	1,314,994
R-squared	0.256	0.257	0.257	0.256	0.257
County-Year	Yes	Yes	Yes	Yes	Yes
Co-Applicant FE	Yes	Yes	Yes	Yes	Yes
Income Bins	Yes	Yes	Yes	Yes	Yes
LTV Bins	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes

Table 10 Banks' S ratings and the origination of mortgage loans. This table presents the regression results in Section 6.4. The dependent variables are the origination of mortgage loans. The key explanatory variables are S ratings and racial groups. All models use annual data for 2018-2019. T-statistics reported in parentheses are based on robust standard errors double-clustered by the bank and by year. All models include County-Year, Co-Applicant, Income Bins, Loan-to-Value (LTV) Bins, and Bank fixed effects. Superscripts ∗∗∗, ∗∗, and ∗ indicate statistical significance at 1%, 5%, and 10% levels.

VARIABLES	(1) LoanApproval	(2) LoanApproval	(3) LoanApproval	(4) LoanApproval	(5) LoanApproval
S	-0.0329	-0.0303	-0.0287	-0.0332	-0.0328
	(0.0555)	(0.0556)	(0.0533)	(0.0571)	(0.0553)
Race x S	0.00894				
	(0.0107)				
AmericanIndian x S		-0.0537**			
		(0.0229)			
Asian x S			0.0178*		
			(0.0102)		
Black x S				-0.00686	
				(0.0158)	
NativeHawaiian x S					-0.152***
					(0.0385)
Race	-0.0519***				
	(0.00769)				
AmericanIndian		-0.0139			
		(0.0136)			
Asian			-0.0407***		
			(0.00721)		
Black				-0.0607***	
				(0.00991)	
NativeHawaiian					0.0331*
					(0.0194)
AUS	0.254***	0.239***	0.240***	0.249***	0.238***
	(0.0510)	(0.0479)	(0.0475)	(0.0507)	(0.0473)
Constant	0.701***	0.712***	0.713***	0.703***	0.715***
	(0.0442)	(0.0430)	(0.0421)	(0.0445)	(0.0427)

Observations	1,811,345	1,532,091	1,667,023	1,641,769	1,525,720
R-squared	0.303	0.297	0.293	0.305	0.297
County-Year	Yes	Yes	Yes	Yes	Yes
Co-Applicant FE	Yes	Yes	Yes	Yes	Yes
Income Bins	Yes	Yes	Yes	Yes	Yes
LTV Bins	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes

Table 11 Banks' S ratings and the amount of mortgage loans. This table presents the regression results in Section 6.4. The dependent variables are the amount of mortgage loans. The key explanatory variables are S ratings and racial groups. All models use annual data for 2018-2019. T-statistics reported in parentheses are based on robust standard errors double-clustered by the bank and by year. All models include County-Year, Co-Applicant, Income Bins, Loan-to-Value (LTV) Bins, and Bank fixed effects. Superscripts ∗∗∗, ∗∗, and ∗ indicate statistical significance at 1%, 5%, and 10% levels.

VARIABLES	(1) LoanAmount	(2) LoanAmount	(3) LoanAmount	(4) LoanAmount	(5) LoanAmount
S	0.0354	0.0464*	0.0379	0.0533**	0.0460*
	(0.0271)	(0.0261)	(0.0272)	(0.0267)	(0.0264)
Race x S	0.00635				
	(0.0141)				
AmericanIndian x S		-0.0685**			
		(0.0277)			
Asian x S			0.0149		
			(0.0157)		
Black x S				-0.0837***	
				(0.0280)	
NativeHawaiian x S					-0.108***
					(0.0350)
Race	-0.0199**				
	(0.00980)				
AmericanIndian		-0.0110			
		(0.0129)			
Asian			0.0493***		
			(0.0104)		
Black				-0.0413***	
				(0.0158)	
NativeHawaiian					0.0333
					(0.0209)
AUS	0.0569***	0.0548***	0.0487***	0.0599***	0.0544***
	(0.0109)	(0.0106)	(0.0108)	(0.0104)	(0.0105)
Constant	12.15***	12.11***	12.16***	12.10***	12.12***
	(0.0211)	(0.0201)	(0.0210)	(0.0202)	(0.0202)

Observations	1,811,345	1,532,091	1,667,023	1,641,769	1,525,720
R-squared	0.790	0.789	0.795	0.786	0.789
County-Year	Yes	Yes	Yes	Yes	Yes
Co-Applicant FE	Yes	Yes	Yes	Yes	Yes
Income Bins	Yes	Yes	Yes	Yes	Yes
LTV Bins	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes

Table 12 Banks' S ratings and interest rate. This table presents the regression results in Section 6.4. The dependent variables are the mortgage loans' interest rates. The key explanatory variables are S ratings and racial groups. All models use annual data for 2018-2019. T-statistics reported in parentheses are based on robust standard errors double-clustered by the bank and by year. All models include County-Year, Co-Applicant, Income Bins, Loan-to-Value (LTV) Bins, and Bank fixed effects. Superscripts ∗∗∗, ∗∗, and ∗ indicate statistical significance at 1%, 5%, and 10% levels.

VARIABLES	(1) InterestRate	(2) InterestRate	(3) InterestRate	(4) InterestRate	(5) InterestRate
S	-0.00326	-0.00382	-0.00338	-0.00362	-0.00372
	(0.00347)	(0.00391)	(0.00368)	(0.00372)	(0.00391)
Race x S	-0.000544				
	(0.000364)				
AmericanIndian x S		-0.00292			
		(0.00202)			
Asian x S			0.000149		
			(0.000357)		
Black x S				-4.00e-05	
				(0.000436)	
NativeHawaiian x S					0.000897
					(0.00197)
Race	6.08e-05				
	(0.000237)				
AmericanIndian		0.00220			
		(0.00155)			
Asian			-0.00116***		
			(0.000270)		
Black				0.000592***	
				(0.000217)	
NativeHawaiian					-0.000242
					(0.00102)
AUS	-0.00308***	-0.00325***	-0.00300***	-0.00320***	-0.00320***
	(0.000931)	(0.000941)	(0.000917)	(0.000946)	(0.000935)
Constant	0.0477***	0.0483***	0.0477***	0.0482***	0.0482***
	(0.00230)	(0.00255)	(0.00242)	(0.00245)	(0.00255)

Observations	1,543,238	1,319,979	1,436,815	1,400,703	1,314,994
R-squared	0.256	0.257	0.257	0.256	0.257
County-Year	Yes	Yes	Yes	Yes	Yes
Co-Applicant FE	Yes	Yes	Yes	Yes	Yes
Income Bins	Yes	Yes	Yes	Yes	Yes
LTV Bins	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes